Multiple Linear Regression – Multi-Channel Marketing Analysis

In [1]:
```
# Importing Libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder, OneHotEncoder, OrdinalEncoder, Stan
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, rand_score, mean_absolute_error, mean_squared
import statsmodels.api as sm
```

In [2]:
```
# Loading dataset
df = pd.read_csv('marketing_sales_data.csv')
# Viewing first 5 rows
df.head()
```

Out[2]:

	TV	Radio	Social Media	Influencer	Sales
0	Low	3.518070	2.293790	Micro	55.261284
1	Low	7.756876	2.572287	Mega	67.574904
2	High	20.348988	1.227180	Micro	272.250108
3	Medium	20.108487	2.728374	Mega	195.102176
4	High	31.653200	7.776978	Nano	273.960377

In [3]:
```
df.columns
```

Out[3]:
```
Index(['TV', 'Radio', 'Social Media', 'Influencer', 'Sales'], dtype='object')
```

Data Exploration and Cleaning

In [4]:
```
# Number of rows and columns
df.shape
```

Out[4]:
```
(572, 5)
```

In [5]:
```
# Information about the data
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 572 entries, 0 to 571
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   TV            572 non-null    object 
 1   Radio         572 non-null    float64
 2   Social Media  572 non-null    float64
 3   Influencer    572 non-null    object 
 4   Sales         572 non-null    float64
dtypes: float64(3), object(2)
memory usage: 22.5+ KB
```

In [6]:
```
# Checking for duplicate records
df.duplicated().sum()
```

Out[6]:
```
np.int64(0)
```

In [7]:
```
# Checking for missing values
df.isna().sum()
```

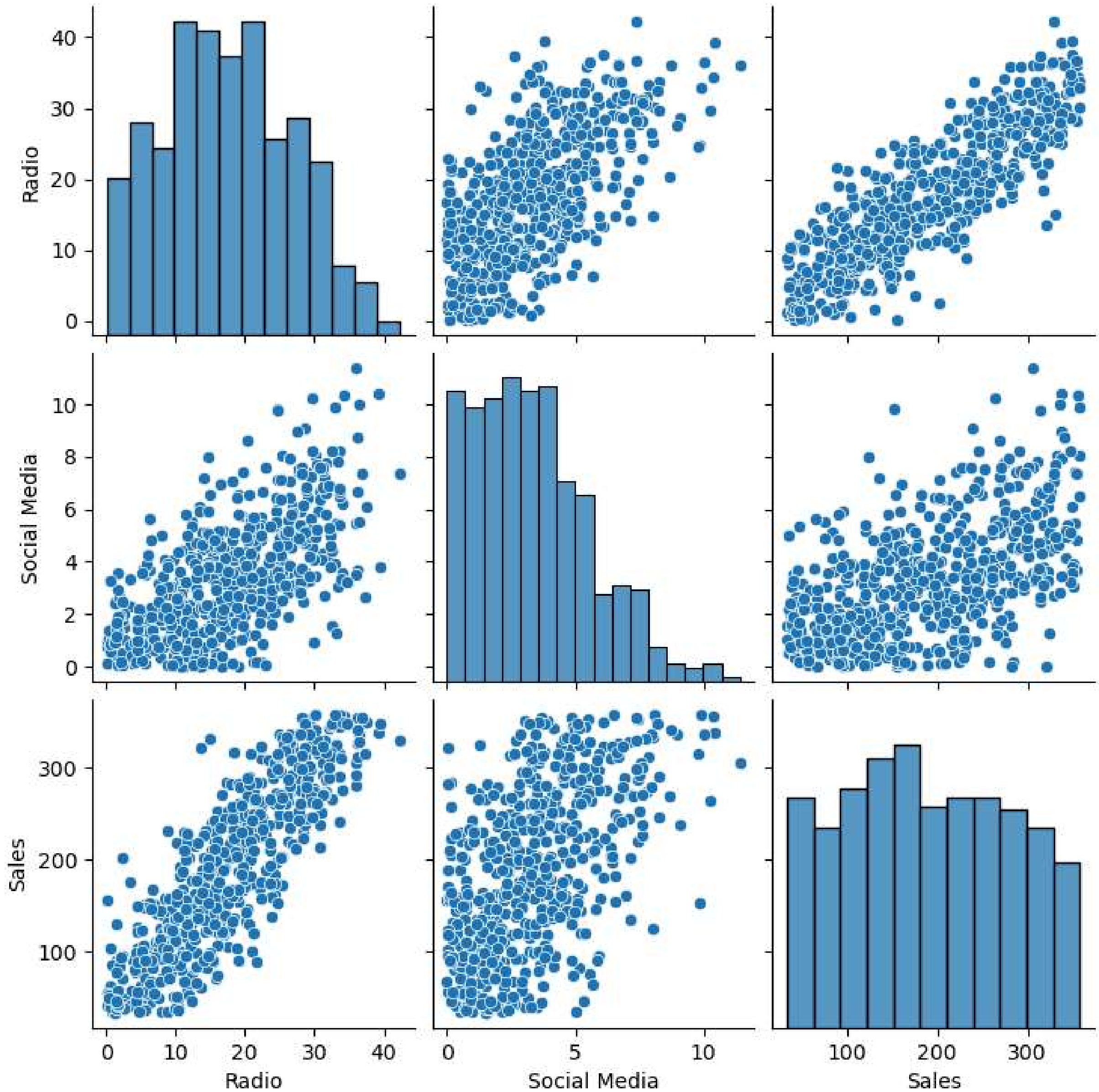
Out[7]:
```
TV              0
Radio           0
Social Media    0
Influencer      0
Sales           0
dtype: int64
```

In [8]:
```
# Summary statistics of numeric variables
df.describe()
```

Out[8]:

	Radio	Social Media	Sales
count	572.000000	572.000000	572.000000
mean	17.520616	3.333803	189.296908
std	9.290933	2.238378	89.871581
min	0.109106	0.000031	33.509810
25%	10.699556	1.585549	118.718722
50%	17.149517	3.150111	184.005362
75%	24.606396	4.730408	264.500118
max	42.271579	11.403625	357.788195

In [9]:
```
# Checking the relationship between variables in the data
sns.pairplot(df)
plt.show()
```



Data Preprocessing and feature scaling

In [10]:

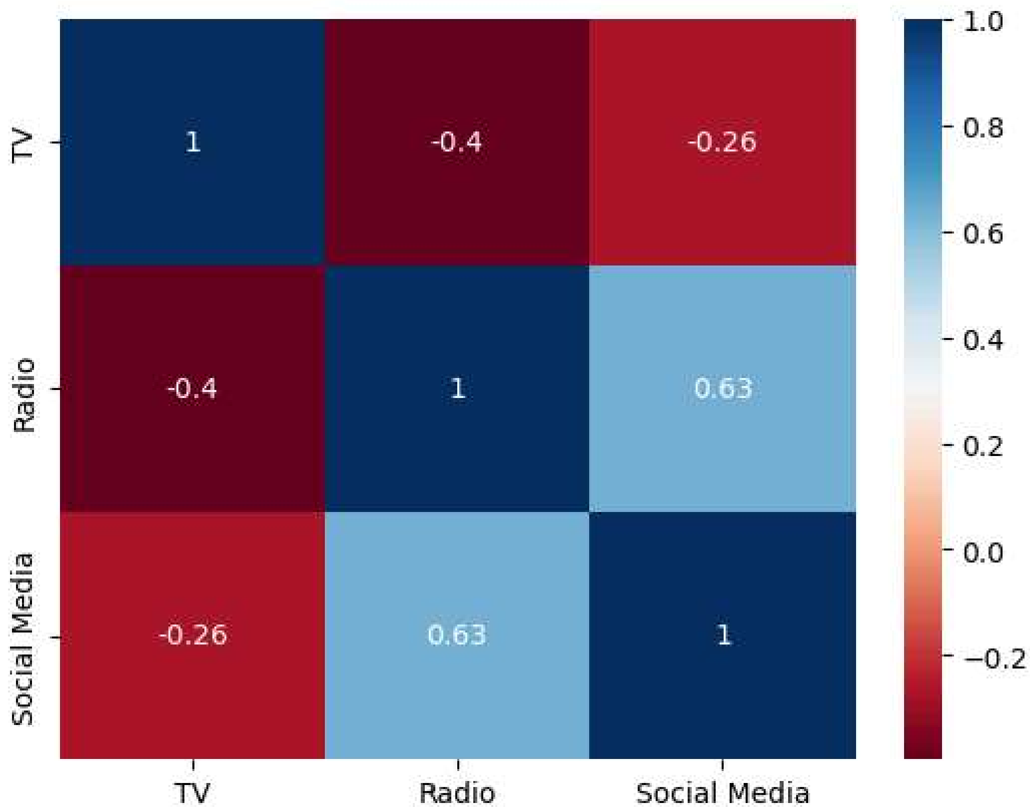
```
# Encoding target variable
label_enc = LabelEncoder()
y = label_enc.fit_transform(df['Sales'])

# Encoding independent variables
num_feat = df[['Radio','Social Media']]
cat_feat = df[['TV','Influencer']]

ord_enc = OrdinalEncoder()
std_scaler = StandardScaler()

cat_feat_encoded = pd.DataFrame(ord_enc.fit_transform(cat_feat),columns=ord_enc.get
num_feat_encoded = pd.DataFrame(std_scaler.fit_transform(num_feat),columns=num_feat

x = pd.concat([cat_feat_encoded,num_feat_encoded],axis=1)
```

In [11]:
```python
x.columns
```

Out[11]:
```
Index(['TV', 'Influencer', 'Radio', 'Social Media'], dtype='object')
```

In [12]:
```python
# Assessing multicollinearity with correlation matrix
var = x[['TV','Radio','Social Media']]
corr = var.corr()
print(corr)
sns.heatmap(corr, annot=True, cmap='RdBu')
plt.show()
```

```
                    TV     Radio  Social Media
TV            1.000000 -0.395841     -0.262081
Radio        -0.395841  1.000000      0.629941
Social Media -0.262081  0.629941      1.000000
```



Building Model with SKLearn

In [13]:
```python
# Splitting dataset - 80% Train; 20% Test
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.2,random_state=42)
```

In [14]:
```python
# Fitting dataset into model for training
lin_reg = LinearRegression()
lin_reg.fit(x_train,y_train)
```

Out[14]:

```
▾ LinearRegression
▶ Parameters
```

In [15]:

```python
lin_reg.intercept_
```

Out[15]:

```
np.float64(296.35055251925615)
```

In [16]:

```python
# Predicting and Evaluating on Test dataset
y_pred = lin_reg.predict(x_test)
mse = mean_squared_error(y_test, y_pred)
rmse = root_mean_squared_error(y_test,y_pred)
mae = mean_absolute_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print(f"Mean Squared Error: {mse:.2f}")
print(f"Root Mean Squared Error: {rmse:.2f}")
print(f"Mean Absolute Error: {mae:.2f}")
print(f"R-Squared calue: {r2:.2f}")
```

```
Mean Squared Error: 9020.50
Root Mean Squared Error: 94.98
Mean Absolute Error: 78.34
R-Squared calue: 0.66
```

Building Model with Statsmodel

In [17]:

```python
x_sm = sm.add_constant(x)

ols_model = sm.OLS(y, x_sm).fit()

print(ols_model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.738
Model:                            OLS   Adj. R-squared:                  0.736
Method:                 Least Squares   F-statistic:                     398.9
Date:                Fri, 19 Jun 2026   Prob (F-statistic):          3.21e-163
Time:                        23:46:55   Log-Likelihood:                -3349.8
No. Observations:                 572   AIC:                             6710.
Df Residuals:                     567   BIC:                             6731.
Df Model:                           4
Covariance Type:            nonrobust
================================================================================
                   coef    std err          t      P>|t|      [0.025      0.975]
--------------------------------------------------------------------------------
const          301.3435      8.185     36.819      0.000     285.268     317.419
TV             -14.1199      4.823     -2.928      0.004     -23.592      -4.648
Influencer      -0.4783      3.200     -0.149      0.881      -6.763       5.806
Radio          137.1026      4.806     28.527      0.000     127.663     146.542
Social Media    -0.1899      4.573     -0.042      0.967      -9.172       8.792
==============================================================================
Omnibus:                        0.177   Durbin-Watson:                   1.859
Prob(Omnibus):                  0.915   Jarque-Bera (JB):                0.092
Skew:                           0.023   Prob(JB):                        0.955
Kurtosis:                       3.043   Cond. No.                         6.05
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spe
cified.
```

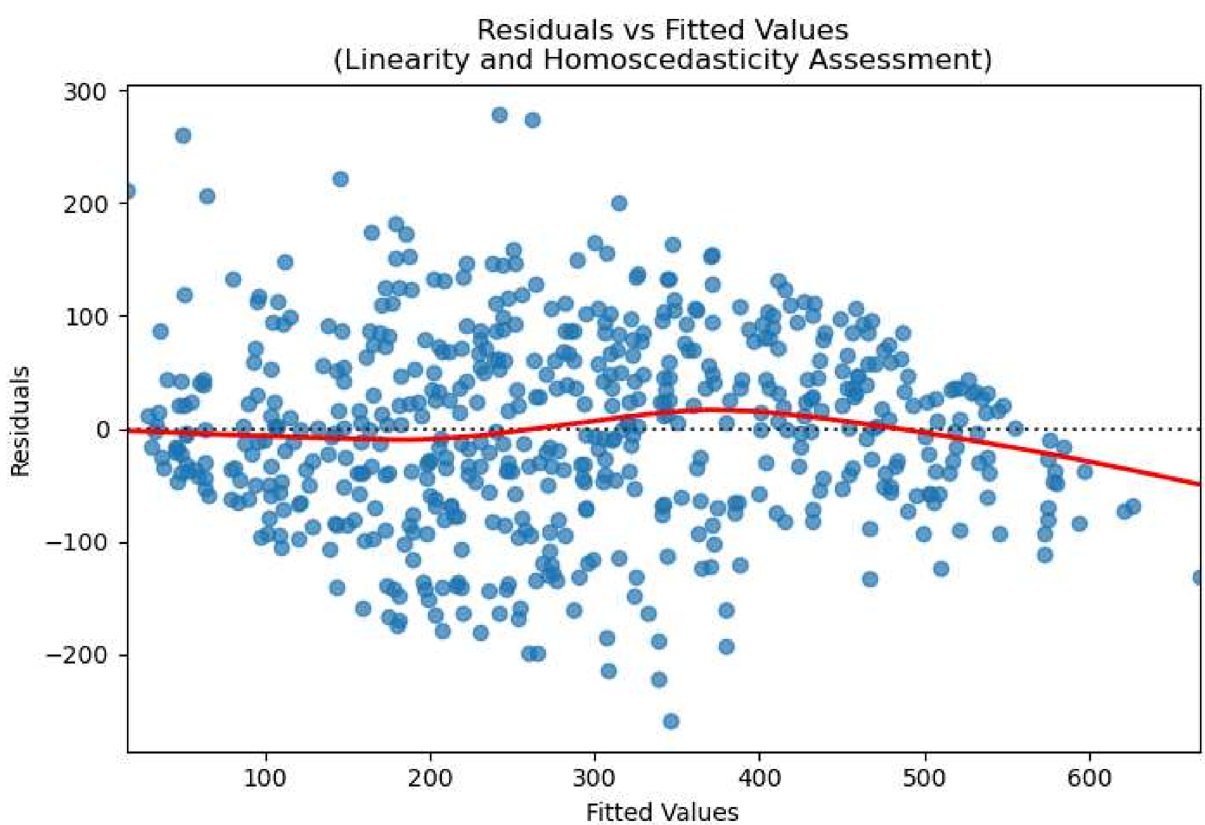
Interpretation of OLS Results

- The multiple linear regression model explains approximately 73.8% of the variation in Sales using TV level, Radio spend, Social Media spend, and Influencer category.
- The adjusted R^2 value of 0.736 indicates that the model retains strong explanatory power after accounting for the number of predictors included.
- The F-test p-value (3.21e-163) results shows that the model results are statistically significant
- Radio advertising is the strongest positive driver of sales in the model. A one-unit increase in Radio advertising spend is associated with an estimated 137.10 unit increase in Sales
- A surprising finding was that a higher TV advertising category is associated with an estimated decrease of 14.12 units in Sales. This could be because TV categories may not represent actual spending amounts
- Influencer type and Social Media spending did not show a statistically significant independent effect on Sales after controlling for other predictors.

In [18]:
```python
# Getting residuals and fitted values
residuals = ols_model.resid
fitted_values = ols_model.fittedvalues
```

Diagnostic plots

In [19]:
```python
# Linearity & Homoscedasticity check
plt.figure(figsize=(8,5))

sns.residplot(
    x=fitted_values,
    y=residuals,
    lowess=True,
    scatter_kws={'alpha':0.7},
    line_kws={'color':'red','linewidth':2}
)

plt.xlabel('Fitted Values')
plt.ylabel('Residuals')

plt.title('Residuals vs Fitted Values\n(Linearity and Homoscedasticity Assessment)'

plt.show()
```

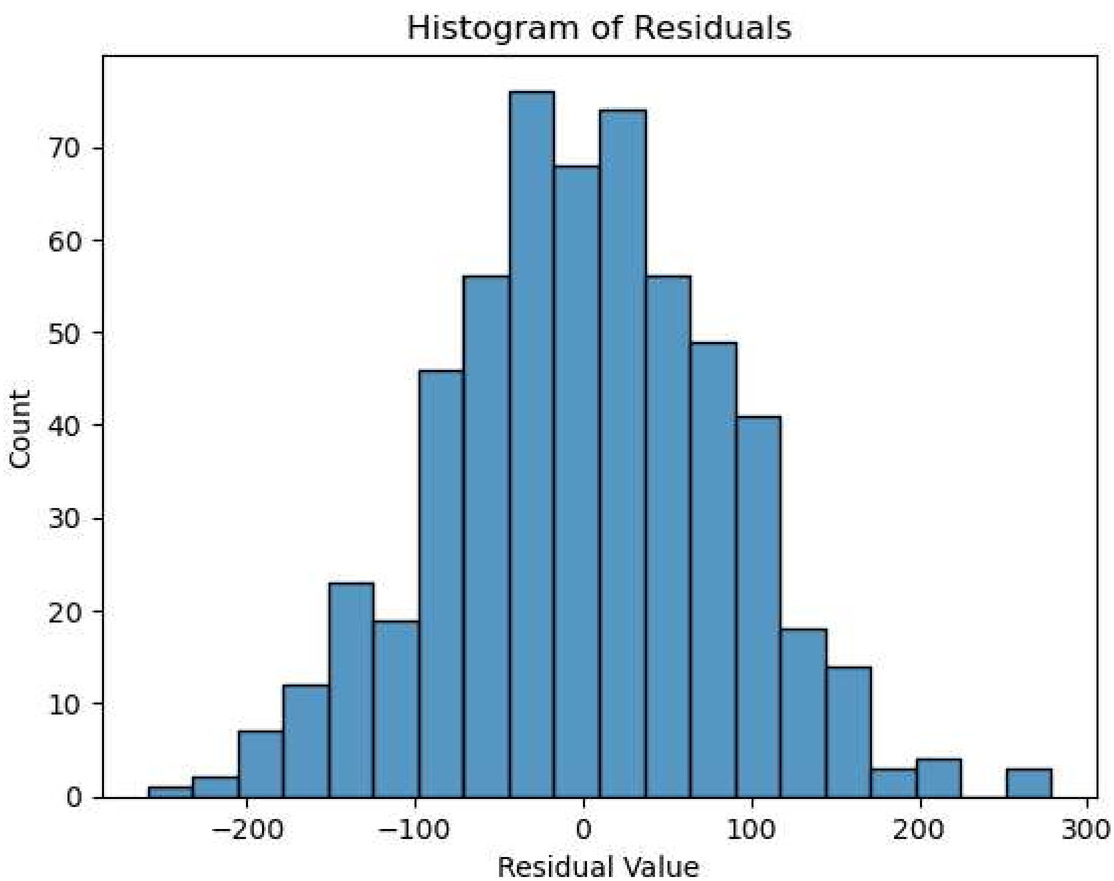


The plot above shows that the assumption of homoscedasticity is reasonably satisfied. However, slight variations in residual spread and the presence of several outliers indicate mild heteroscedasticity.

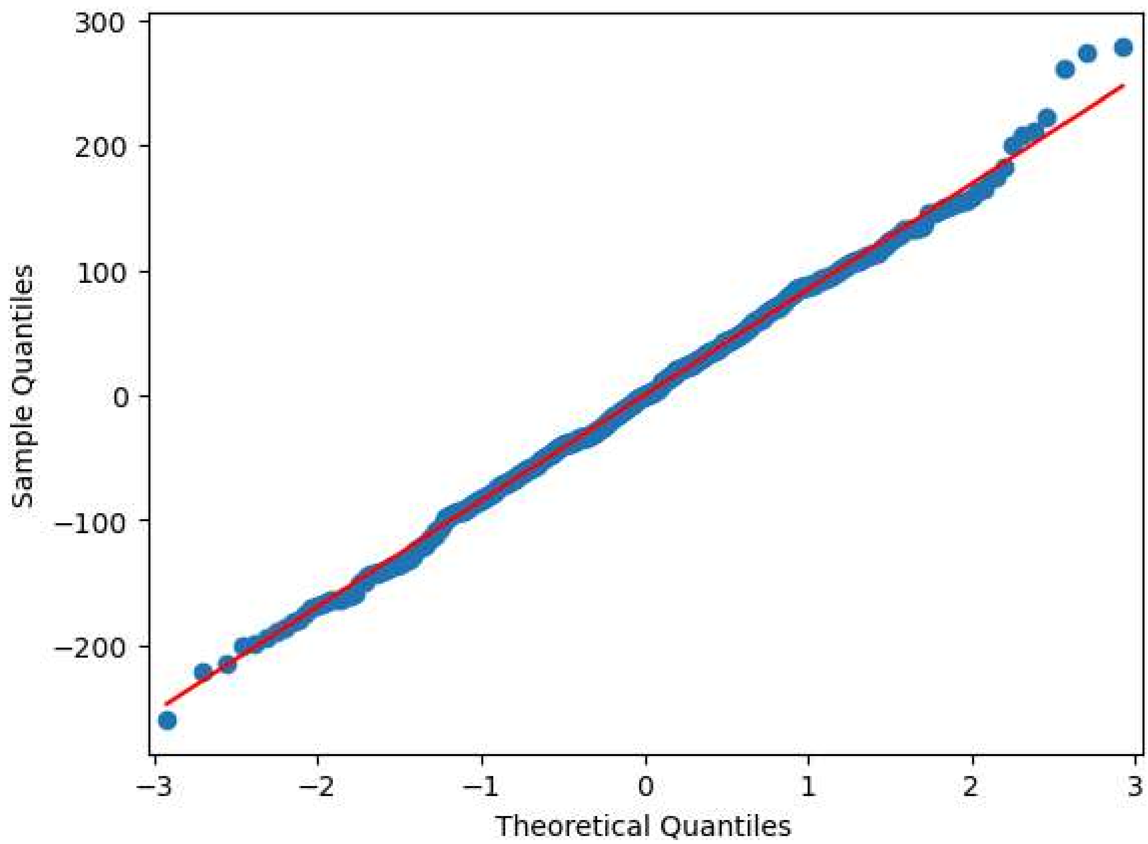
The LOWESS line showed only slight curvature at higher fitted values, indicating minor deviations from perfect linearity.

Nonetheless, both assumptions were met.

In [20]:
```
# Checking Normality Assumption
fig = sns.histplot(residuals)
fig.set_xlabel("Residual Value")
fig.set_title("Histogram of Residuals")
plt.show()
```



In [21]:
```
# Q-Q Plot

fig = sm.qqplot(ols_model.resid, line = 's')
plt.show()
```



Omnibus p = 0.915

Jarque-Bera p = 0.955

The histogram, Q-Q plot, and the values above reveal that the residuals are normally distributed

Recommendations

- The regression analysis shows that Radio advertising is the strongest driver of Sales, with a statistically significant positive relationship.
- The company should prioritize investment in Radio marketing channels while continuously evaluating the effectiveness of TV and Social Media campaigns.
- Although Social Media and Influencer variables were included in the model, they did not demonstrate significant independent contributions to sales, suggesting that their impact may depend on other factors or require further analysis.